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OPTICAL SYSTEM

Abstract

Disclosed is an optical system including a display device, an optical element, and an optical shutter. The optical element is located over the display device and is configured to transmit light from the display device. The optical shutter is located over the optical element. The optical element is further configured to maintain a polarization axis of the light in a driving state and rotate the polarization axis of the light by 90° in a non-driving state. The optical shutter has a pair of linear polarizers with transmission axes intersecting each other.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority to Japanese Patent Application No. 2024-037452, filed on Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] An embodiment of the present invention relates to an optical system having an image-displaying function. For example, an embodiment of the present invention relates to an optical system capable of capturing an image created with light having different polarization states and allowing the image to be viewed.

BACKGROUND

[0003] An in-vehicle display device in which a TN (Twisted Nematic) liquid crystal element is arranged on a liquid crystal display has been known (Japanese laid-open patent publication No. 2006-208606). Since the polarization axis of the light from the liquid crystal display device can be rotated by the TN liquid crystal element in this display device, it is possible to change the polarization axis of the light forming an image by appropriately driving the TN liquid crystal element. As a result, the use of the difference in reflection characteristics between light with different polarization axes not only prevents the reflection of images on reflective surfaces such as windshields but also allows images to be displayed on the reflective surfaces by utilizing the reflection.

SUMMARY

[0004] An embodiment of the present invention is an optical system. The optical system includes a display device, an optical element, and an optical shutter. The optical element is located over the display device and is configured to transmit light from the display device. The optical shutter is located over the optical element. The optical element is further configured to maintain a polarization axis of the light in a driving state and rotate the polarization axis of the light by 90° in a non-driving state. The optical shutter has a pair of linear polarizers with transmission axes intersecting each other.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0005] FIG. 1 is a schematic developed perspective view of an optical system according to an embodiment of the present invention.

[0006] FIG. 2 is a schematic top view of a display device of an optical system according to an embodiment of the present invention.

[0007] FIG. 3 is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

[0008] FIG. 4 is a schematic side view of an optical system according to an embodiment of the present invention.

[0009] FIG. 5 is a schematic perspective view for explaining an operation of an optical system according to an embodiment of the present invention.

[0010] FIG. 6 is a schematic perspective view for explaining an operation of an optical system according to an embodiment of the present invention.

[0011] FIG. 7 is a schematic perspective view for explaining an operation of an optical system according to an embodiment of the present invention.

[0012] FIG. 8 is a schematic perspective view for explaining an operation of an optical system according to an embodiment of the present invention.

[0013] FIG. 9 is a schematic perspective view for explaining an operation of an optical system according to an embodiment of the present invention.

[0014] FIG. **10** is a schematic view for explaining an operation of an optical system according to an embodiment of the present invention.

[0015] FIG. **11** is a schematic view for explaining an operation of an optical system according to an embodiment of the present invention.

[0016] FIG. **12** is a schematic side view for explaining an operation of an optical system according to an embodiment of the present invention.

[0017] FIG. **13** is a schematic side view for explaining an operation of an optical system according to an embodiment of the present invention.

[0018] FIG. **14** is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

[0019] FIG. **15** is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

[0020] FIG. **16** is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

[0021] FIG. **17** is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

[0022] FIG. **18** is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

[0023] FIG. **19** is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

[0024] FIG. **20** is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

[0025] FIG. **21** is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

[0026] FIG. **22** is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

[0027] FIG. **23** is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

[0028] FIG. **24** is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

[0029] FIG. **25** is a schematic cross-sectional view of a portion of an optical system according to an embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

[0030] Hereinafter, each embodiment of the present invention is explained with reference to the drawings. The invention can be implemented in a variety of different modes within its concept and should not be interpreted only within the disclosure of the embodiments exemplified below.

[0031] The drawings may be illustrated so that the width, thickness, shape, and the like are illustrated more schematically compared with those of the actual modes in order to provide a clearer explanation. However, they are only an example, and do not limit the interpretation of the invention. In the specification and the drawings, the same reference number is provided to an element that is the same as that which appears in preceding drawings, and a detailed explanation may be omitted as appropriate. The reference number is used when plural structures which are the same as or similar to each other are collectively represented, while a hyphen and a natural number are further used when these structures are independently represented.

[0032] In the specification and the claims, unless specifically stated, when a state is expressed where a structure is arranged “over” another structure, such an expression includes both a case where the substrate is arranged immediately above the “other structure” so as to be in contact with the “other structure” and a case where the structure is arranged over the “other structure” with an additional structure therebetween.

[0033] In the specification and the claims, an expression that two structures “orthogonally

intersect” or “are perpendicular to each other” includes not only a state where the two structures intersect each other at 90° but also a state where the two structures intersect each other at an angle of $90^\circ \pm 10^\circ$. An expression that “two structures are parallel” includes a state where an angle between extending directions of the two structures is $0^\circ \pm 10^\circ$.

First Embodiment

[0034] In this embodiment, an optical system according to an embodiment of the present invention is explained.

1. Structure

[0035] FIG. 1 is a schematic developed perspective view of an optical system **100** according to an embodiment of the present invention. As shown in FIG. 1, the optical system **100** includes a display device **110** and an optical element **200** provided over the display device **110** and overlapping the display device **110**. As described below in detail, a backlight **102** is further provided when the display device **110** includes a liquid crystal element. Note that, although the components such as the display device **110**, the optical element **200**, and the backlight **102** are separately illustrated in FIG. 1, these components are fixed to one another directly or by means of adhesive layers, fixing jigs, housings, and the like. For example, the display device **110** and the optical element **200** may be fixed to each other with an adhesive layer **104** containing a polymer such as an epoxy resin and an acrylic resin. Although not illustrated in FIG. 1, the optical system **100** further includes an optical shutter as described below.

(1) Backlight

[0036] The backlight **102** is a light source for irradiating the optical system **100** with light. Since a known structure can be employed for the backlight **102**, a detailed description is omitted. Briefly, the backlight **102** has a cold cathode tube and/or a light-emitting diode (LED) serving as a light source and is composed of a reflector for effective use of the light from the light source, a light diffuser for uniformly diffusing the light, a light guide plate, a prism sheet, and the like. The light from the backlight **102** is incident on the optical element **200** through the display device **110**.

(2) Display Device

[0037] As the display device **110**, a liquid crystal display device and an electroluminescence display device can be utilized. The display device **110** may be an active-matrix type display device or a passive-matrix type display device. As an example, an active-matrix type liquid crystal display device is used as the display device **110** in this embodiment.

[0038] FIG. 2 shows a schematic top view of the display device **110**, and FIG. 3 shows a schematic cross-sectional view of the optical system **100** excluding the optical shutter. As shown in these drawings, the display device **110** has a substrate **112** and a counter substrate **124** opposing the substrate **112**. The substrate **112** and the counter substrate **124** are components for transmitting the light from the display device **110** and for providing mechanical strength to the display device **110**. Accordingly, both the substrate **112** and the counter substrate **124** are configured to transmit visible light and include glass or a resin such as a polyimide and a polycarbonate. The substrate **112** and the counter substrate **124** may be flexible.

[0039] A variety of patterned insulating films, conductive films, and semiconductor films are stacked between the substrate **112** and the counter substrate **124**. These films constitute pixels **120**, driver circuits (scanning-line driver circuit **114** and signal-line driver circuit **116**), and terminals **118** as well as pixel circuits including a transistor **130**. Electric power and a variety of signals are input from an external circuit, which is not illustrated, through the terminals **118**, and control signals (gate signals, initialization signals, video signals, and so on) for displaying images are generated by the driver circuits and are supplied to the pixels **120**. Control of the pixels with the control signals allows the display device **100** to display images.

[0040] There is no restriction on the structure of the pixel circuit, and the pixel circuit is composed of a plurality of transistors including the transistor **130** and one or a plurality of capacitance elements which are not illustrated. In the example shown here, the transistor **130** is provided over

an undercoat **122**, which is an inorganic insulating film, and is composed of a semiconductor film **132**, a gate insulating film **134**, a gate electrode **136**, an interlayer insulating film **138**, a pair of terminals **140** and **142**, and the like. The configuration of the transistor **130** can also be arbitrarily determined, and the transistor **130** may be a top-gate type transistor shown in FIG. 3, a bottom-gate type transistor, or the like. A leveling film **144** is provided over the pixel circuits to absorb the unevenness thereof.

[0041] The display device **110** further includes a liquid crystal element **150** over the leveling film **144**. The liquid crystal element **150** shown in FIG. 3 is a so-called TN liquid crystal element and is composed of a pixel electrode **152** electrically connected to the transistor **130**, a first orientation film **154** over the pixel electrode **152**, a liquid crystal layer **156** over the first orientation film **154**, a second orientation film **158** over the liquid crystal layer **156**, a common electrode **160** to which a common potential is applied, and the like. Over the common electrode **160**, a color filter **172** overlapping the pixel electrode **152** through an overcoat **174**, which is an optional component, and a black matrix **170** for preventing light leakage between adjacent pixels **120** are provided. These configurations allow the pixel **120** to function as the smallest unit providing color information. Although not illustrated, the liquid crystal element **150** is not limited to a TN liquid crystal element, and a VA (Vertical Alignment) liquid crystal element or an IPS (In-Plane Switching) liquid crystal element may be utilized.

[0042] The first orientation film **154** and the second orientation film **158** include a resin such as a polyimide and are configured to orient liquid crystal molecules included in the liquid crystal layer **156** in a certain direction. Hence, the first orientation film **154** and the second orientation film **158** are subjected to an alignment process such as a rubbing process or are formed using photo-alignment. The directions in which the first orientation film **154** and the second orientation film **158** respectively orient the liquid crystal molecules (which is the direction of the long axis of the liquid crystal molecules when they are oriented under the influence of the orientation film, and is, hereinafter, referred to as an orientation direction) are orthogonal to each other.

[0043] The display device **110** is further provided with a first linear polarizer **180** under the substrate **112** and a second linear polarizer **182** over the counter substrate **124**. The first linear polarizer **180** and the second linear polarizer **182** are arranged to exist in a crossed Nicols relationship with each other. Accordingly, the light from the backlight **102** becomes linearly polarized light (e.g., P-polarized light) when passing through the first linear polarizer **180**. When the liquid crystal element **150** is a TN liquid crystal element, this linearly polarized light is optically rotated by 90° by the liquid crystal layer **156** to become linearly polarized light (e.g., S-polarized light) and passes through the second linear polarizer **182** when the display device **110** is not driven. Therefore, the linearly polarized light is incident on the optical element **200**.

[0044] The transmission axis of the first linear polarizer **180** may be parallel to a side of the substrate **112** or may be inclined from a side of the substrate **112**. In the latter case, it is preferable to provide the first linear polarizer **180** so that the transmission axis of the first linear polarizer **180** is inclined 45° with respect to a side of the substrate **112**. Since the first linear polarizer **180** and the second linear polarizer **182** are in a crossed Nicols relationship as described above, when the transmission axis of the first linear polarizer **180** is parallel to a side of the substrate **112**, the transmission axis of the second linear polarizer **182** is perpendicular to this side. On the other hand, when the transmission axis of the first linear polarizer **180** is inclined with respect to a side of the substrate **112**, the transmission axis of the second linear polarizer **182** is also inclined at the same angle with respect to this side.

(3) Optical Element

[0045] The optical element **200** is a component for transmitting the light emitted from the display device **110** and for controlling the polarization axis thereof and includes a TN liquid crystal element. Specifically, as shown in FIG. 1 and FIG. 3, the optical element **200** has a substrate **202** and a counter substrate **204** opposing the substrate **202**. Similar to the substrate **112** and the counter

substrate **124**, the substrate **202** and the counter substrate **204** are configured to transmit the light from the display device **110** and to provide mechanical strength to the optical element **200**. Therefore, the substrate **202** and the counter substrate **204** are also configured to transmit visible light and include glass or a resin such as a polyimide and a polycarbonate. The substrate **202** and/or the counter substrate **204** may also be flexible.

[0046] A first electrode **212**, a third orientation film **214** over the first electrode **212**, a liquid crystal layer **216** over the third orientation film **214**, a fourth orientation film **218** over the liquid crystal layer **216**, and a second electrode **220** over the fourth orientation film **218** are provided between the substrate **202** and the counter substrate **204**. The TN liquid crystal element is composed of the first electrode **212**, the third orientation film **214**, the liquid crystal layer **216**, the fourth orientation film **218**, and the second electrode **220**. As an optional component, the optical element **200** may have a protective insulating film **210** and/or a protective insulating film **222** between the substrate **202** and the first electrode **212** and between the counter substrate **204** and the second electrode **220**, respectively. The protective insulating films **210** and **222** may be composed of one or a plurality of films containing a silicon-containing inorganic compound such as silicon nitride and silicon oxide. It is possible to prevent impurities contained in the substrate **202** and the counter substrate **204**, such as alkali metals, from entering the liquid crystal layer **216** by the protective insulating films **210** and **222**.

[0047] Both the first electrode **212** and the second electrode **220** are configured to transmit visible light. Therefore, the first electrode **212** and the second electrode **220** preferably include a conductive oxide transmitting visible light, such as indium-tin oxide (ITO) and indium-zinc oxide (IZO). The first electrode **212** and the second electrode **220** may each be formed as a single conductive film, or, although not illustrated, may be composed of a plurality of conductive films arranged in a stripe form. For example, one of the first electrode **212** and the second electrode **220** may be formed as a single conductive film overlapping all of the pixels **120**, and the other may be formed as a plurality of conductive films arranged in a stripe form.

[0048] Similar to the first orientation film **154** and the second orientation film **158**, the third orientation film **214** and the fourth orientation film **218** also contain a resin such as a polyimide and are configured to orient nematic liquid crystal molecules included in the liquid crystal layer **216** in a certain direction. Therefore, the third orientation film **214** and the fourth orientation film **218** are subjected to an alignment process such as a rubbing process or are formed by using photo-alignment. The orientation directions of the third orientation film **214** and the fourth orientation film **218** are also orthogonal to each other. Furthermore, the third orientation film **214** is preferably provided so that the orientation direction is parallel to the transmission axis of the second linear polarizer **182**. As described below, arrangement of the optical element **200** with such a configuration allows the polarization axis of the linearly polarized light incident on the optical element **200** from the display device **110** to be rotated by 90° by the liquid crystal layer **216** when the optical element **200** is not driven and to be maintained when the optical element **200** is driven.

[0049] In the optical system **100** having the aforementioned configuration, it is not necessary to arrange any polarizer in the optical element **200**. Therefore, the adhesive layer **104** securing the display device **110** and the optical element **200** to each other may be in direct contact with the second linear polarizer **182** and the substrate **202**. In addition, the counter substrate **204** may be directly exposed to the outside air.

(4) Optical Shutter

[0050] As shown in FIG. 4, the optical system **100** has an optical shutter **230** over the optical element **200**. The optical shutter **230** includes a pair of linear polarizers **232** and **234** with transmission axes orthogonal to each other. In order to distinguish the polarizers **232** and **234** from the first linear polarizer **180** and the second linear polarizer **182**, the linear polarizers **232** and **234** are hereinafter referred to as the polarizers for optical selection **232** and **234**, respectively. The polarizer for optical selection **232** is positioned parallel to the display device **110**. That is, the

polarizer for optical selection **232** is provided so that the main surface thereof is orthogonal to the normal of the main surface of the display device **110** (e.g., the main surface of the substrate **112**). On the other hand, the other polarizer for optical selection **234** is arranged so as to be inclined with respect to the display device **110**. Specifically, the polarizer for optical selection **234** is provided so that the normal of the main surface thereof is inclined from the normal of the main surface of the display device **110** at a polar angle $\theta_{\text{sub.1}}$. The polar angle $\theta_{\text{sub.1}}$ is, for example, $20^\circ \pm 5^\circ$. [0051] Furthermore, the optical shutter **230** is configured to selectively or preferentially transmit the light emitted from the optical element **200** through one of the polarizers for optical selection **232** and **234** when the optical element **200** is driven and to selectively or preferentially transmit this light through the other of the polarizers for optical selection **232** and **234** when the optical element **200** is not driven. For example, the polarizer for optical selection **232** is arranged so that its transmission axis (see the straight arrow in the drawing) is parallel to the polarization axis of the light emitted from the optical element **200** when the optical element **200** is not driven. In the optical system **100** having the aforementioned configuration, since the light is optically rotated by 90° by the optical element **200** in the non-driving state, the polarization axis of the light, which is emitted from the optical element **200** and is incident on the polarizer for optical selection **232** when the optical element **200** is not driven, is perpendicular to the transmission axis of the second linear polarizer **182**. Therefore, the polarizer for optical selection **232** is arranged so that its transmission axis is parallel to the transmission axis of the second linear polarizer **182**. On the other hand, since the transmission axis of the polarizer for optical selection **234** is orthogonal to that of the polarizer for optical selection **232**, the polarizer for optical selection **234** is arranged so that its transmission axis is perpendicular to the transmission axis of the second linear polarizer **182**. With this configuration, when the optical element **200** is not driven, the image displayed by the display device **110** can be selectively or preferentially acquired through the polarizer for optical selection **234**. In contrast, when the optical element **200** is driven, the polarization axis of the light incident on the optical element **200** is maintained so that an image can be selectively or preferentially acquired through the polarizer for optical selection **232**.

[0052] Conversely, the optical system **100** may be configured so that an image is acquired through the polarizer for optical selection **232** when the optical element **200** is not driven and so that an image is acquired through the polarizer for optical selection **234** when the optical element **200** is driven. In this case, the polarizer for optical selection **232** may be arranged so that its transmission axis is parallel to the polarization axis of the light emitted from the optical element **200** when the optical element **200** is not driven. That is, the polarizer for optical selection **232** may be arranged so that its transmission axis is perpendicular to the transmission axis of the second linear polarizer **182**. Conversely, since the transmission axis of the polarizer for optical selection **234** is orthogonal to that of the polarizer for optical selection **232**, the polarizer for optical selection **234** is arranged so that its transmission axis is parallel to the transmission axis of the second linear polarizer **182**. Hence, it is possible to view and acquire images formed by the light with different polarization states by using the optical system **100** having the above configuration.

2. Optical Control by Optical System

[0053] As described above, the optical element **200** disposed over the display device **110** is a TN liquid crystal element. Therefore, when the optical element **200** is not driven, that is, when no potential difference is provided between the first electrode **212** and the second electrode **220**, the liquid crystal molecules are oriented according to the orientation directions of the third orientation film **214** and the fourth orientation film **218**. Specifically, the liquid crystal molecules are oriented along the orientation direction of the third orientation film **214** on the first electrode **212** side and along the orientation direction of the fourth orientation film **218** on the second electrode **220** side. Since the orientation directions of the third orientation film **214** and the fourth orientation film **218** are orthogonal to each other, the orientation direction of the liquid crystal molecules twists as they approach the second electrode **220** from the first electrode **212**, eventually twisting by 90° .

Accordingly, the light incident on the optical element **200** optically rotates by 90° within the liquid crystal layer **216**. In contrast, when the optical element **200** is driven, that is, when a potential difference is provided between the first electrode **212** and the second electrode **220** to generate a sufficient longitudinal electric field therebetween, the liquid crystal molecules are oriented in the direction of the electric field, i.e., in the direction perpendicular to the first electrode **212**. In this case, the light incident on the optical element **200** does not optically rotate, and the polarization axis is maintained.

[0054] When the transmission axis of the second linear polarizer **182** is parallel to a side of the substrate **112**, the polarization axis of the light emitted from the display device **110** (white arrow in the drawing) is parallel to the transmission axis of the second linear polarizer **182** (solid arrow) as shown in FIG. 5. When the optical element **200** is not driven (OFF), the polarization axis of the light emitted from the optical element **200** becomes perpendicular to the transmission axis of the second linear polarizer **182** since the polarization axis of the incident light is rotated by 90° by the optical element **200**. On the other hand, when the optical element **200** is driven (ON), the polarization axis of the light emitted from optical element **200** becomes parallel to the transmission axis of the second linear polarizer **182** (see white arrow in FIG. 6) because the polarization axis of the incident light is not rotated.

[0055] The same is applied when the transmission axis of the second linear polarizer **182** is inclined from a side of the substrate **112**. Specifically, the polarization axis of the light emitted from the display device **110** (white arrow) is inclined from a side of the substrate **112** as shown in FIG. 7 by arranging the second linear polarizer **182** so that the transmission axis thereof is inclined from a side of the substrate **112**. When the optical element **200** is not driven (OFF), the polarization axis of the light emitted from the optical element **200** is also inclined from this side because the polarization axis of the incident light is rotated by 90° by the optical element **200**. When the polarization axis of the light emitted from the display device **110** is inclined 45° from a side of the substrate **112**, the polarization axis of the light emitted from the optical element **200** is also inclined 135° from this side. On the other hand, when the optical element **200** is driven (ON), the polarization axis of the light emitted from the optical element **200** (FIG. 8, white arrow) is also inclined from this side because the polarization axis of the incident light is maintained without rotation. When the polarization axis of the light emitted from the display device **110** is inclined 45° from a side of the substrate **112**, the polarization axis of the light emitted from the optical element **200** is also inclined 45° from this side.

[0056] The application of these features described above allows images formed by the light with different polarization states to be acquired and viewed depending on the angle from which the optical system **100** is viewed (i.e., polar angle). This point is explained using FIG. 9 to FIG. 11. In the following explanation, the case in which the polarization axis of the polarized light emitted from the display system **110** is inclined 45° from a side of the substrate **202** as shown in FIG. 9 is used as an example. The directions of two adjacent sides of the substrate **202** are defined as an x-direction and a y-direction, respectively, and the normal direction of the substrate **202** is defined as a z-direction. The polar angle $\theta_{\text{sub.1}}$ is an angle from the z-direction to the y direction, and an azimuth $\theta_{\text{sub.2}}$ is an angle from the x-direction to the y-direction.

[0057] When the display device **110** is driven, red, green, or blue light with controlled gradation is obtained from each pixel **120**, and these lights are combined to form an image. This image is formed by the light having a polarization axis inclined by 45° from a side of the substrate **112** or the substrate **202** and is incident on the optical element **200**. Here, as schematically shown in FIG. 10, the polarization state of the light observed through the optical element **200** at a polar angle $\theta_{\text{sub.1}}$ of 0° (i.e., frontal to the optical system **100**) is almost the same when the optical element **200** is driven and is not driven, and is almost independent on the azimuth $\theta_{\text{sub.2}}$. Specifically, the Stokes parameter $S_{\text{sub.3}}$ is almost 0 even when the azimuth $\theta_{\text{sub.2}}$ changes and is independent on the azimuth $\theta_{\text{sub.2}}$ so that the light emitted from the optical system **100** is almost linearly

polarized.

[0058] Even when the polar angle $\theta_{\text{sub.1}}$ deviates from 0° and an image is observed at an angle inclined from the normal of the optical system **100** (e.g., when the polar angle $\theta_{\text{sub.1}}$ is) 20° , linearly polarized light is observed regardless of the azimuth $\theta_{\text{sub.2}}$ as shown in FIG. **11** when the optical element **200** is not driven. In contrast, when the optical element **200** is driven, the polarization state is highly dependent on the azimuth $\theta_{\text{sub.2}}$. Although the light emitted from the optical system **100** is almost linearly polarized when the azimuth $\theta_{\text{sub.2}}$ is approximately 45° , 135° , 225° , and 315° , the light deviates from linear polarization and approaches circular polarization at other angles. In particular, when the azimuth $\theta_{\text{sub.2}}$ is approximately 0° , 90° , 180° , and 270° , the light emitted from the optical system **100** is closest to circular polarized light and becomes elliptically polarized light. That is, the intensity difference between mutually orthogonal polarized lights becomes maximum.

[0059] Therefore, it is possible to observe and acquire images formed by the light with different polarization states according to the polar angle $\theta_{\text{sub.1}}$ by switching the optical element **200** on and off in the optical system **100**. As described above, since the polarizer for optical selection **232** of the optical shutter **230** is provided so as to be parallel to the display device **110** (see FIG. **4**), an image with a polar angle $\theta_{\text{sub.1}}$ of 0° is incident on the polarizer for optical selection **232**. As shown in FIG. **12**, when the optical element **200** is not driven, the polarization axis of the light incident on the optical element **200** from the display device **110** is rotated by 90° by the optical element **200**. Therefore, this image is shielded by the polarizer for optical selection **232** by arranging the polarizer for optical selection **232** so that the polarization axis of the light emitted from the optical element **200** (white arrow) is orthogonal to the transmission axis of the polarizer for optical selection **232** (see solid arrow). On the other hand, the image is incident on the polarizer for optical selection **234** at the polar angle $\theta_{\text{sub.1}}$. Furthermore, the transmission axis of the polarizer for optical selection **234** is orthogonal to the transmission axis of the polarizer for optical selection **232** as described above. Therefore, the light emitted from the optical element **200** can pass through the polarizer for optical selection **234** by arranging the polarizer for optical selection **234** so that the light emitted from the optical element **200** is incident at the azimuth $\theta_{\text{sub.2}}$ at which the Stokes parameter $S_{\text{sub.3}}$ is minimum or maximum. Therefore, when the optical element **200** is not driven, the image passing through the polarizer for optical selection **234** can be selectively or preferentially acquired.

[0060] On the other hand, when the optical element **200** is driven, the polarization axis of the light incident on the optical element **200** is maintained even after passing through the optical element **200** (see the white arrow and the solid arrow thereunder in FIG. **13**) so that the polarization axis of this light is parallel to the transmission axis of the polarizer for optical selection **232** (solid arrow). Thus, the image can be acquired through the polarizer for optical selection **232**. On the other hand, on the polarizer for optical selection **232** arranged at the apolar angle $\theta_{\text{sub.1}}$, the light having a polarization axis perpendicular to its transmission axis is incident. Therefore, the light emitted through the optical element **200** is shielded by the polarizer for optical selection **234**. This mechanism allows the image passing through the polarizer for optical selection **232** to be selectively acquired when the optical element **200** is driven.

[0061] In view of the purpose of observing orthogonal polarization states at different polar angles, when considering which polarization state between the driving state and the non-driving state is more advantageous, it is preferable to use the non-driven state when the polar angle $\theta_{\text{sub.1}}$ is not 0 degrees and the driven state when the polar angle $\theta_{\text{sub.1}}$ is 0 degrees.

[0062] Note that it is not always necessary to incline the polarizer for optical selection **234** from the display device **110**, and the polarizer for optical selection **234** may be arranged parallel to the display device **110**. In this case, it is preferable to arrange the polarizer for optical selection **234** so that the light passing through the optical element **200** is incident at the Brewster angle at which the reflectance of one of the polarized lights is 0.

[0063] As described above, the use of the optical system **100** allows images formed by the light with different polarization states to be selectively acquired and viewed. That is, it is possible to selectively acquire and view images formed by the light with linearly polarized light with polarization axes orthogonal to each other. Although not illustrated, the images respectively acquired through the polarizers for optical selection **232** and **234** may be superimposed using a half mirror, polarization beam splitter, or the like by which a plurality of images can be synthesized using a single display device **110**.

Second Embodiment

[0064] In this embodiment, modified examples of the optical system **100** described in the First Embodiment are explained. An explanation of the structures the same as or similar to those described in the First Embodiment may be omitted. The following modified examples can be combined with the optical system **100** described in the First Embodiment as appropriate, and the modified examples may also be combined with each other.

1. Modified Example 1

[0065] In the optical system **100** described in the First Embodiment, the pair of linear polarizers (first linear polarizer **180** and second linear polarizer **182**) is provided to the display device **110**. The optical system **100** according to the Modified Example 1 may further be provided with a half-wave plate ($\frac{1}{2}$ wave plate). For example, the display device **110** may have a $\frac{1}{2}$ wave plate **184** over the second linear polarizer **182** as shown in FIG. **14**. In this case, the adhesive layer **104** securing the display device **110** and the optical element **200** may be in direct contact with the substrate **202** and the $\frac{1}{2}$ wave plate **184**. Alternatively, the optical element **200** may have the $\frac{1}{2}$ wave plate **184** under the substrate **202** as shown in FIG. **15**. In this case, the adhesive layer **104** may be in direct contact with the second linear polarizer **182** and the $\frac{1}{2}$ wave plate **184**.

[0066] In the Modified Example 1, the $\frac{1}{2}$ wave plate **184** rotates the polarization axis of the light emitted from the display device **110** by 45° . Thus, when the polarization axis of the light emitted from the display device **110** is parallel to a side of the substrate **202**, for example, this light can be incident on the optical element **200** after converting this light to the polarized light with a polarization axis inclined by 45° with respect to this side. Conversely, when the polarization axis of the light emitted from the display device **110** is inclined 45° from a side of the substrate **202**, this light can be incident on the optical element **200** after converting this light to the polarized light having a polarization axis parallel to this side.

2. Modified Example 2

[0067] In the optical system **100**, a third linear polarizer **186** may further be disposed between the substrate **202** and the counter substrate **124**. For example, the third linear polarizer **186** may be provided under the substrate **202** (FIG. **16**). In this case, the adhesive layer **104** may be in direct contact with the third linear polarizer **186** and the second linear polarizer **182**. The third linear polarizer **186** is arranged so that the transmission axis thereof is parallel to that of the second linear polarizer **182**.

[0068] As described above, the display **110** and the optical element **200** are fixed to each other by the adhesive layer **104**. However, if there is a misalignment during the fixing process, the polarization axis of the light emitted from the display **110** is not necessarily parallel to a side of the substrate **202** or inclined at a predetermined angle (for example,) 45° from this side. However, even if such misalignment occurs, the polarization axis of the light incident on the optical element **200** can be set parallel to a side of the substrate **202** or at a predetermined angle from this side by providing the third linear polarizer **186**.

[0069] Alternatively, the third linear polarizer **186** may not be provided to the optical element **200** but may be provided to the display device **110** (FIG. **17**). In this case, the third linear polarizer **186** is provided over the second linear polarizer **182**. The adhesive layer **104** may be in direct contact with the third linear polarizer **186** and the substrate **202**.

3. Modified Example 3

[0070] In the optical system **100** described in the First Embodiment, the pair of linear polarizers (first linear polarizer **180** and second linear polarizer **182**) is provided to the display device **110**. In contrast, in the optical system **100** according to the Modified Example 3, although the first linear polarizer **180** is provided under the substrate **112**, the second linear polarizer **182** is not provided to the display device **110** as shown in FIG. **18**. The second linear polarizer **182** is provided under the substrate **202** of the optical element **200**, by which gradation based on the video signals can be obtained from each pixel **120**. The adhesive layer **104** may be in direct contact with the counter substrate **124** and the second linear polarizer **182**.

[0071] In Modified Example 3, the light incident on the TN liquid crystal element responsible for the optical rotation becomes linearly polarized light through the second linear polarizer **182** located under the substrate **202** so that the switching of the polarization state of the light forming images can also be performed by the optical element **200**. Although not illustrated, the third linear polarizer **186** may also be provided in the Modified example 3, similar to the Modified Example 2. The third linear polarizer **186** is provided between the second linear polarizer **182** and the substrate **202**. The third linear polarizer **186** may be in direct contact with the second linear polarizer **182**.

[0072] Similar to the Modified Example 1, the $\frac{1}{2}$ wave plate **184** may also be provided in the Modified Example 3. For example, the optical element **200** may have the second linear polarizer **182** and the $\frac{1}{2}$ wave plate **184** sandwiched between the second linear polarizer **182** and the substrate **202** as shown in FIG. **19**.

4. Modified Example 4

[0073] In the display device **110** of the aforementioned optical systems **100**, the liquid crystal element **150** is used as the display element. However, the configuration of the display device **110** is not limited thereto, and an electroluminescence element **190** may be used as the display element as shown in FIG. **20**. The electroluminescence element **190** is composed of a pixel electrode **152**, a common electrode **160** over the pixel electrode **152**, and an electroluminescence layer **192** therebetween. An insulating bank **146** containing a polymer such as a polyimide and a polysiloxane resin is provided at the edge of the pixel electrode **152**, which prevents the electroluminescence layer **192** from being cut and electrically insulates the adjacent pixels **120**. The electroluminescence layer **192** may be composed of a plurality of functional layers containing an organic compound. Known configurations may be applied to the electroluminescence layer **192**, and the functional layers such as a charge-injection layer, a charge-transporting layer, a charge-blocking layer, and an emission layer may be combined as appropriate. As an optional component, a sealing film **194** may be provided over the common electrode **160** to protect the electroluminescence element **190**. When the electroluminescence element **190** is used as the display element, the backlight **102** is not necessary.

[0074] In the display device **110** of the optical system **100** according to the Modified Example 4, it is not necessary to provide the first linear polarizer **180**, and the second linear polarizer **182** may be provided over the counter substrate **124**. This configuration allows the light emitted from the electroluminescence element **190** to be converted to linearly polarized light and to be supplied to the optical element **200**. Alternatively, the second linear polarizer **182** may be provided under the substrate **202** of the optical element **200** without providing the second linear polarizer **182** to the display device **110** (FIG. **21**). Alternatively, the third linear polarizer **186** may be placed over the second linear polarizer **182** of the display device **110** in order to prevent defects caused by the misalignment described above (FIG. **22**). Alternatively, the second linear polarizer **182** may be placed over the counter substrate **124** of the display device **110**, and the third linear polarizer **186** may further be placed under the substrate **202** of the optical element **200** (FIG. **23**).

5. Modified Example 5

[0075] Similar to the Modified Example 1, the $\frac{1}{2}$ wave plate **184** may also be used in the case where the electroluminescence element **190** is used as the display element. For example, the second linear polarizer **182** and the $\frac{1}{2}$ wave plate **184** may be sequentially arranged over the counter

substrate **124** of the display device **110** as shown in FIG. **24**. Alternatively, no polarizer may be provided to the display **110**, while the second linear polarizer **182** may be placed under the substrate **202** of the optical element **200**, and the $\frac{1}{2}$ wave plate **184** may be placed between the substrate **202** and the second linear polarizer **182** as shown in FIG. **25**.

[0076] Similar to the optical system **100** described in the First Embodiment, it is possible to selectively acquire and view images formed by mutually orthogonal linearly polarized lights in any of the aforementioned Modified Examples. It is also possible to synthesize a plurality of images using a single display device **110**.

[0077] The aforementioned modes described as the embodiments of the present invention can be implemented by appropriately combining with each other as long as no contradiction is caused. Furthermore, any mode which is realized by persons ordinarily skilled in the art through the appropriate addition, deletion, or design change of elements or through the addition, deletion, or condition change of a process on the basis of each embodiment is included in the scope of the present invention as long as they possess the concept of the present invention.

[0078] It is understood that another effect different from that provided by each of the aforementioned embodiments is achieved by the present invention if the effect is obvious from the description in the specification or readily conceived by persons ordinarily skilled in the art.

Claims

1. An optical system comprising: a display device; an optical element located over the display device and configured to transmit light from the display device; and an optical shutter located over the optical element configured to maintain a polarization axis of the light in a driving state and rotate the polarization axis by 90° in a non-driving state, wherein the optical shutter comprises a pair of linear polarizers with transmission axes intersecting each other.
2. The optical system according to claim 1, wherein the display device comprises: a first linear polarizer; a liquid crystal element over the first linear polarizer; and a second linear polarizer over the liquid crystal element, and the optical element comprises a TN liquid crystal element.
3. The optical system according to claim 2, further comprising a $\frac{1}{2}$ wave plate between the display device and the optical element.
4. The optical system according to claim 1, wherein the display element comprises: a first linear polarizer; and a liquid crystal element over the first linear polarizer, wherein the optical element comprises: a second linear polarizer; and a TN liquid crystal element over the second linear polarizer.
5. The optical system according to claim 4, further comprising a $\frac{1}{2}$ wave plate between the second linear polarizer and the TN liquid crystal element.
6. The optical system according to claim 1, wherein the display device comprises: an electroluminescence display device; and a first linear polarizer over the electroluminescence display device, and the optical element comprises a TN liquid crystal element.
7. The optical system according to claim 6, further comprising a $\frac{1}{2}$ wave plate between the electroluminescence display device and the optical element.
8. The optical system according to claim 1, wherein the display device comprises an electroluminescence display device, and the optical element comprises: a first linear polarizer; and a TN liquid crystal element over the first linear polarizer.
9. The optical system according to claim 8, further comprising a $\frac{1}{2}$ wave plate between the first linear polarizer and the TN liquid crystal element.
10. The optical system according to claim 2, further comprising a third linear polarizer between the display device and the optical element, wherein a transmission axis of the third linear polarizer is parallel to the polarization axis of the light from the display device.
11. The optical system according to claim 4, further comprising a third linear polarizer between the

display device and the optical element, wherein a transmission axis of the third linear polarizer is parallel to the polarization axis of the light from the display device.

12. The optical system according to claim 6, further comprising a third linear polarizer between the display device and the optical element, wherein a transmission axis of the third linear polarizer is parallel to the polarization axis of the light from the display device.

13. The optical system according to claim 7, further comprising a third linear polarizer between the display device and the optical element, wherein a transmission axis of the third linear polarizer is parallel to the polarization axis of the light from the display device.

14. The optical system according to claim 1, wherein the pair of linear polarizers comprises a first polarizer for optical selection and a second polarizer for optical selection, the first polarizer for optical selection is arranged parallel to the display device, and the second polarizer for optical selection is arranged so that a normal thereof inclines from a normal of a main surface of the display device.

15. The optical system according to claim 14, wherein the normal of the main surface of the second polarizer for optical selection and the normal of the main surface of the display device incline from each other by $20^{\circ} \pm 5^{\circ}$.

16. The optical system according to claim 14, wherein the first polarizer for optical selection is arranged so that a transmission axis thereof is parallel to a polarization axis of light from the optical element in a non-driving state of the optical element.
